Math 54 — Week 11

Symmetric matrices and the singular value decomposition (Lay 7.1, 7.4); second-order linear equations (Nagle 4.2–4.3)

§1 Facts used from Weeks 1–10

(W1) **Diagonalization** (Week 8). $A = PDP^{-1}$ with D diagonal exactly when A has n independent eigenvectors (the columns of P). The eigenvalues are the roots of $\det(A - \lambda I)$, and for each eigenvalue, the dimension of its eigenspace is at most its multiplicity as a root. Similar matrices ($A = PBP^{-1}$) have the same characteristic polynomial (Week 7).

(W2) **Orthogonal matrices** (Week 9). A square U with orthonormal columns has $U^\mathsf{T}U = I$ and $U^{-1} = U^\mathsf{T}$. A product of orthogonal matrices is orthogonal (Week 9, P7). Entry (i, j) of $A^\mathsf{T}B$ is column i of A times column j of B; a matrix with orthonormal columns has $U^\mathsf{T}U = I$ even if it is not square (Week 9, Thm. 6). Products can be computed in blocks, and a product XY equals the sum over j of (column j of X)(row j of Y) (Lay 2.4).

(W3) **Gram–Schmidt** (Week 10). Every subspace of $\mathbb{R}^n$ has an orthonormal basis. An orthonormal set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ in $\mathbb{R}^n$ extends to an orthonormal basis of $\mathbb{R}^n$: extend it to a basis (Week 6) and apply Gram–Schmidt, which leaves $\mathbf{u}_1, \dots, \mathbf{u}_k$ unchanged because they are already orthonormal.

(W4) **Complex numbers** (Week 8). For $z = a + bi$, $\bar{z} = a - bi$ and $z\bar{z} = a^2 + b^2 \geq 0$. A complex number is 0 exactly when its real and imaginary parts are 0. Every polynomial of degree n has n complex roots counting multiplicity (the fundamental theorem of algebra, cited). A complex root λ of $\det(A - \lambda I)$ has an eigenvector $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{C}^n$ (Week 8, Lay 5.5), and for a real matrix A, the conjugate of $A\mathbf{x}$ is $A\bar{\mathbf{x}}$.

(W5) **Subspaces** (Weeks 6, 9). $\operatorname{rank} A + \dim \operatorname{Nul} A = n$; $(\operatorname{Col} A)^\perp = \operatorname{Nul} A^\mathsf{T}$ and $(\operatorname{Row} A)^\perp = \operatorname{Nul} A$; $\dim W + \dim W^\perp = n$ and $(W^\perp)^\perp = W$; $\mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} = \|A\mathbf{x}\|^2$.

(W6) **Projections** (Weeks 9–10). For an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_p$ of W, $\operatorname{proj}_W \mathbf{y} = UU^\mathsf{T}\mathbf{y}$ with $U = [\,\mathbf{u}_1 \cdots \mathbf{u}_p\,]$; $\mathbf{u}\mathbf{u}^\mathsf{T}$ is the projection matrix onto $\operatorname{Span}\{\mathbf{u}\}$ for a unit vector $\mathbf{u}$. Least-squares solutions are the $\hat{\mathbf{x}}$ with $A\hat{\mathbf{x}} = \operatorname{proj}_{\operatorname{Col} A} \mathbf{b}$.

(W7) **Calculus.** $\frac{d}{dt}e^{rt} = re^{rt}$ for real r; the product rule; the Maclaurin series of e^x, $\cos x$, $\sin x$; $\cos(x - y) = \cos x \cos y + \sin x \sin y$.

(W8) **The mass–spring model** (Nagle 4.1, cited). The position $y(t)$ of a mass m on a spring of stiffness k with damping coefficient b satisfies $my'' + by' + ky = 0$ when no outside force acts.

§2 This week

§2.1 Symmetric matrices (Lay 7.1)

The question. The symmetric matrix

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

has eigenvectors $(1, 1)$ (eigenvalue 4) and $(1, -1)$ (eigenvalue 2), and these are orthogonal. Is that an accident? Can every symmetric matrix be diagonalized with an orthogonal P, so that $P^{-1} = P^\mathsf{T}$ needs no computation?

Definition. A is **symmetric** if $A^\mathsf{T} = A$ (so A is square and $a_{ij} = a_{ji}$). A is **orthogonally diagonalizable** if $A = PDP^\mathsf{T}$ with P an orthogonal matrix and D diagonal.

Theorem (Eigenvectors of a symmetric matrix (Lay Thm. 1))
If A is symmetric, then eigenvectors from different eigenspaces are orthogonal.

The idea

For a symmetric A, A can be moved from one side of a dot product to the other: $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$. Compute $(A\mathbf{v}_1) \cdot \mathbf{v}_2$ both ways and compare.

Proof. 1. For any $\mathbf{x}, \mathbf{y}$, $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^\mathsf{T} A^\mathsf{T} \mathbf{y} = \mathbf{x}^\mathsf{T} A\mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$, because $A^\mathsf{T} = A$.

2. Let $A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$ and $A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$ with $\lambda_1 \neq \lambda_2$. By step 1,

$$\lambda_1(\mathbf{v}_1 \cdot \mathbf{v}_2) = (A\mathbf{v}_1) \cdot \mathbf{v}_2 = \mathbf{v}_1 \cdot (A\mathbf{v}_2) = \lambda_2(\mathbf{v}_1 \cdot \mathbf{v}_2).$$

3. So $(\lambda_1 - \lambda_2)(\mathbf{v}_1 \cdot \mathbf{v}_2) = 0$, and since $\lambda_1 \neq \lambda_2$, $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$.

□

Theorem (The spectral theorem (Lay Thms. 2 and 3))
An $n \times n$ matrix A is orthogonally diagonalizable if and only if it is symmetric. A symmetric A has these properties:

(a) A has n real eigenvalues, counting multiplicities;

(b) the dimension of each eigenspace equals the multiplicity of its eigenvalue;

(c) eigenspaces of different eigenvalues are orthogonal to each other;

(d) A is orthogonally diagonalizable.

The idea

One direction is a transpose: $(PDP^\mathsf{T})^\mathsf{T} = PDP^\mathsf{T}$. For the other, first show the eigenvalues are real by computing $\bar{\mathbf{x}}^\mathsf{T} A\mathbf{x}$ for a complex eigenvector, which is real when A is real and symmetric. Then use one unit eigenvector $\mathbf{u}_1$: in an orthonormal basis that starts with $\mathbf{u}_1$, the matrix of A splits into λ_1 and a smaller symmetric block, and induction handles the block.

Proof. 1. *Orthogonally diagonalizable implies symmetric.* If $A = PDP^\mathsf{T}$, then $A^\mathsf{T} = (P^\mathsf{T})^\mathsf{T} D^\mathsf{T} P^\mathsf{T} = PDP^\mathsf{T} = A$, because $D^\mathsf{T} = D$.

2. *(a) The eigenvalues are real.* Let λ be a complex root of $\det(A - \lambda I)$, with an eigenvector $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{C}^n$ (W4), and let $q = \bar{\mathbf{x}}^\mathsf{T} A\mathbf{x}$. Conjugating (with A real) and then transposing the 1×1 matrix gives

$$\bar{q} = \mathbf{x}^\mathsf{T} A\bar{\mathbf{x}} = (\mathbf{x}^\mathsf{T} A\bar{\mathbf{x}})^\mathsf{T} = \bar{\mathbf{x}}^\mathsf{T} A^\mathsf{T} \mathbf{x} = \bar{\mathbf{x}}^\mathsf{T} A\mathbf{x} = q,$$

so q is real. Also $q = \lambda\, \bar{\mathbf{x}}^\mathsf{T} \mathbf{x}$, and $\bar{\mathbf{x}}^\mathsf{T} \mathbf{x} = |x_1|^2 + \cdots + |x_n|^2$ is real and positive (W4). So $\lambda = q/(\bar{\mathbf{x}}^\mathsf{T} \mathbf{x})$ is real. By (W4), $\det(A - \lambda I)$ has n roots counting multiplicity, all real.

3. *(d), by induction on n.* For $n = 1$, $A = [a] = [1][a][1]$. Suppose every symmetric $(n-1) \times (n-1)$ matrix is orthogonally diagonalizable. Take a real eigenvalue λ_1 of A (step 2) with a unit eigenvector $\mathbf{u}_1 \in \mathbb{R}^n$, and extend to an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ (W3). Let $U = [\,\mathbf{u}_1\ W\,]$ with $W = [\,\mathbf{u}_2 \cdots \mathbf{u}_n\,]$.

4. In blocks (W2),

$$U^\mathsf{T} AU = \begin{bmatrix} \mathbf{u}_1^\mathsf{T} A\mathbf{u}_1 & \mathbf{u}_1^\mathsf{T} AW \\ W^\mathsf{T} A\mathbf{u}_1 & W^\mathsf{T} AW \end{bmatrix}.$$

Since $A\mathbf{u}_1 = \lambda_1 \mathbf{u}_1$: the corner is $\lambda_1 \mathbf{u}_1^\mathsf{T} \mathbf{u}_1 = \lambda_1$, and $W^\mathsf{T} A\mathbf{u}_1 = \lambda_1 W^\mathsf{T} \mathbf{u}_1 = \mathbf{0}$ because $\mathbf{u}_2, \ldots, \mathbf{u}_n$ are orthogonal to $\mathbf{u}_1$. Also $\mathbf{u}_1^\mathsf{T} AW = (A^\mathsf{T} \mathbf{u}_1)^\mathsf{T} W = (A\mathbf{u}_1)^\mathsf{T} W = \mathbf{0}$. And $B = W^\mathsf{T} AW$ has $B^\mathsf{T} = W^\mathsf{T} A^\mathsf{T} W = B$. So

$$U^\mathsf{T} AU = \begin{bmatrix} \lambda_1 & 0 \\ 0 & B \end{bmatrix}, \qquad B = W^\mathsf{T} AW \text{ symmetric, } (n-1) \times (n-1).$$

5. By induction $B = VD'V^{\mathsf{T}}$ with V orthogonal. Let

$$V_1 = \begin{bmatrix} 1 & 0 \\ 0 & V \end{bmatrix}, \qquad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & D' \end{bmatrix}.$$

V_1 is orthogonal (its columns are orthonormal because those of V are), and multiplying in blocks,

$$V_1 D V_1^{\mathsf{T}} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & VD'V^{\mathsf{T}} \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & B \end{bmatrix} = U^{\mathsf{T}} A U.$$

Since $U^{-1} = U^{\mathsf{T}}$, multiplying by U on the left and U^{T} on the right gives

$$A = (UV_1)\, D\, (UV_1)^{\mathsf{T}},$$

where UV_1 is orthogonal (W2). So A is orthogonally diagonalizable; this also proves the "if" direction.

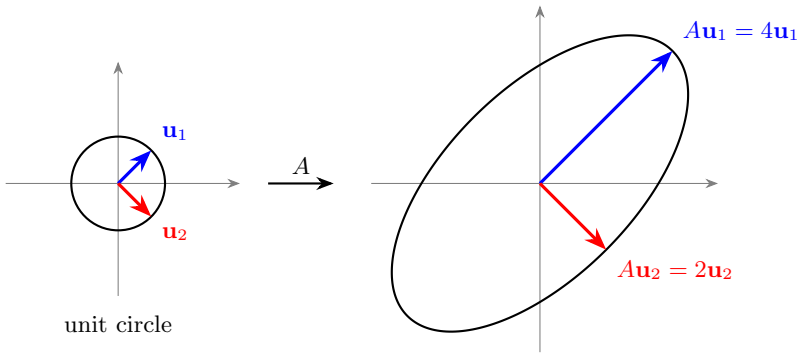
6. *(b).* With $A = PDP^{\mathsf{T}} = PDP^{-1}$, A and D are similar, so $\det(A - \lambda I) = \det(D - \lambda I)$ (W1), and the multiplicity of λ is the number of times λ appears on the diagonal of D. The corresponding columns of P are that many orthonormal eigenvectors for λ, so the eigenspace has dimension at least the multiplicity; by (W1) it is at most the multiplicity.

7. *(c)* is Thm. 1.

□

The question answered. Yes, by the spectral theorem. For the matrix of the question, normalize the eigenvectors:

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}, \qquad A = PDP^{\mathsf{T}}.$$



A symmetric matrix stretches along perpendicular axes. The matrix A of the question maps the unit circle to an ellipse whose axes point along the orthonormal eigenvectors $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)$ and $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$, with half-lengths 4 and 2: a unit vector $\mathbf{x} = \cos\theta\,\mathbf{u}_1 + \sin\theta\,\mathbf{u}_2$ goes to $A\mathbf{x} = 4\cos\theta\,\mathbf{u}_1 + 2\sin\theta\,\mathbf{u}_2$.

Example (a repeated eigenvalue). Orthogonally diagonalize

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

1. $A(1,1,1) = (4,4,4)$, so 4 is an eigenvalue with eigenvector $(1,1,1)$. $A - I$ has all entries 1, so it has rank 1, and $\lambda = 1$ has the 2-dimensional eigenspace $x_1 + x_2 + x_3 = 0$. By (b), the multiplicities 1 and 2 account for all three eigenvalues.

2. The basis $(1,-1,0)$, $(1,0,-1)$ of the $\lambda = 1$ eigenspace is not orthogonal. Gram–Schmidt inside the eigenspace:

$$(1,0,-1) - \frac{1}{2}(1,-1,0) = \Big(\frac{1}{2}, \frac{1}{2}, -1\Big), \quad \text{use } (1,1,-2).$$

3. Both are orthogonal to $(1,1,1)$ by Thm. 1. Normalize:

$$P = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad A = PDP^{\mathsf{T}}.$$

Spectral decomposition. Expanding $A = (PD)P^{\mathsf{T}}$ as a sum of (column)(row) products (W2),

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^{\mathsf{T}} + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^{\mathsf{T}} + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^{\mathsf{T}}.$$

Reason: $PD = [\,\lambda_1 \mathbf{u}_1 \cdots \lambda_n \mathbf{u}_n\,]$, and row j of P^{T} is $\mathbf{u}_j^{\mathsf{T}}$. Each $\mathbf{u}_j \mathbf{u}_j^{\mathsf{T}}$ is the projection matrix onto $\mathrm{Span}\{\mathbf{u}_j\}$ (W6) and has rank 1 (every column is a multiple of $\mathbf{u}_j$), so A is a weighted sum of projections onto perpendicular lines. For the 2×2 example,

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = 4 \cdot \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 2 \cdot \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Shortcuts

- **Eigenvectors for different eigenvalues need no Gram–Schmidt**: they are already orthogonal (Thm. 1); only normalize. Use Gram–Schmidt only inside an eigenspace of dimension 2 or more. Instance: the 3×3 example.
- 2×2 **symmetric**: once one eigenvector (a, b) for one eigenvalue is known, an eigenvector for the other eigenvalue is $(-b, a)$, by Thm. 1 (the other eigenspace is a line orthogonal to (a, b)). Instance: $(1, 1)$ gives $(-1, 1)$ for the matrix of the question.
- **Inverse of** P: $P^{-1} = P^{\mathsf{T}}$, so $A = PDP^{\mathsf{T}}$ needs no row reduction. Reason: (W2). Instance: for the question, $P^{\mathsf{T}} = P$, since that P is symmetric.
- **Last eigenspace by orthogonality**: for a 3×3 symmetric matrix with eigenvalues λ_1 (simple) and λ_2 (double), the λ_2 eigenspace is the plane orthogonal to the λ_1 eigenvector, by (b) and (c). Instance: the plane $x_1 + x_2 + x_3 = 0$ above.

Traps

- Eigenvectors in the same eigenspace need not be orthogonal: $(1, -1, 0)$ and $(1, 0, -1)$ are both eigenvectors for $\lambda = 1$ above, with dot product 1.
- The columns of P must be unit vectors. With the unnormalized eigenvectors $(1, 1)$, $(1, -1)$ as the columns of P, $P^{\mathsf{T}}P = 2I$, so $PDP^{\mathsf{T}} = 2A$, not A.
- An orthogonal matrix need not be orthogonally diagonalizable: the rotation by 90°, with columns $(0, 1)$ and $(-1, 0)$, is orthogonal but not symmetric, and it has no real eigenvalues at all.
- A diagonalizable matrix that is not symmetric is not orthogonally diagonalizable. This matrix has the eigenvectors $(1, 0)$ and $(1, 1)$, which are not orthogonal:

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}.$$

§2.2 The singular value decomposition (Lay 7.4)

The question. The matrix

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

is not symmetric, but it still maps the unit circle to an ellipse. How long are the axes of the ellipse, in which directions do they point, and which unit vectors $\mathbf{x}$ make $\|A\mathbf{x}\|$ largest?

The length $\|A\mathbf{x}\|$ is determined by the matrix $A^\mathsf{T}A$, which is symmetric because $(A^\mathsf{T}A)^\mathsf{T} = A^\mathsf{T}A$, and $\|A\mathbf{x}\|^2 = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x}$ (W5). By the spectral theorem, $A^\mathsf{T}A$ has an orthonormal basis of eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with real eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Each $\lambda_i \geq 0$, because

$$\lambda_i = \lambda_i\, \mathbf{v}_i \cdot \mathbf{v}_i = \mathbf{v}_i^\mathsf{T} A^\mathsf{T} A \mathbf{v}_i = \|A\mathbf{v}_i\|^2.$$

Definition. The **singular values** of an $m \times n$ matrix A are $\sigma_i = \sqrt{\lambda_i}$, where $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ are the eigenvalues of $A^\mathsf{T}A$. By the computation above, $\sigma_i = \|A\mathbf{v}_i\|$.

The largest stretch is σ_1. For a unit vector $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ (so $c_1^2 + \cdots + c_n^2 = 1$),

$$\|A\mathbf{x}\|^2 = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} = \lambda_1 c_1^2 + \cdots + \lambda_n c_n^2 \leq \lambda_1(c_1^2 + \cdots + c_n^2) = \lambda_1,$$

using $A^\mathsf{T}A\mathbf{x} = \sum \lambda_i c_i \mathbf{v}_i$ and orthonormality. Equality holds for $\mathbf{x} = \mathbf{v}_1$. So the maximum of $\|A\mathbf{x}\|$ over unit vectors is σ_1, reached at $\mathbf{v}_1$. In the same way, for unit vectors orthogonal to $\mathbf{v}_1$ (so $c_1 = 0$) the bound is λ_2: the maximum of $\|A\mathbf{x}\|$ over those is σ_2, reached at $\mathbf{v}_2$.

Theorem (The images $A\mathbf{v}_i$ (Lay Thm. 9))

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be an orthonormal basis of eigenvectors of $A^\mathsf{T}A$ with $\lambda_1 \geq \cdots \geq \lambda_n$, and suppose A has exactly r nonzero singular values. Then $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ is an orthogonal basis of $\operatorname{Col} A$, and $\operatorname{rank} A = r$.

Proof. 1. For $i \neq j$: $(A\mathbf{v}_i) \cdot (A\mathbf{v}_j) = \mathbf{v}_i^\mathsf{T} A^\mathsf{T} A \mathbf{v}_j = \lambda_j\, \mathbf{v}_i \cdot \mathbf{v}_j = 0$. So the $A\mathbf{v}_i$ are orthogonal.

2. $\|A\mathbf{v}_i\| = \sigma_i$, which is nonzero for $i \leq r$ and zero for $i > r$. So $A\mathbf{v}_1, \dots, A\mathbf{v}_r$ are nonzero and orthogonal, hence independent (Week 9, Thm. 4).

3. They span $\operatorname{Col} A$: every $\mathbf{y} = A\mathbf{x}$ in $\operatorname{Col} A$, with $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, is

$$\mathbf{y} = c_1 A\mathbf{v}_1 + \cdots + c_n A\mathbf{v}_n = c_1 A\mathbf{v}_1 + \cdots + c_r A\mathbf{v}_r,$$

since $A\mathbf{v}_i = \mathbf{0}$ for $i > r$. So they are a basis of $\operatorname{Col} A$, and $\operatorname{rank} A = \dim \operatorname{Col} A = r$.

□

Theorem (Singular value decomposition (Lay Thm. 10))

Let A be $m \times n$ of rank r, with nonzero singular values $\sigma_1 \geq \cdots \geq \sigma_r$. Then

$$A = U\Sigma V^\mathsf{T},$$

where U $(m \times m)$ and V $(n \times n)$ are orthogonal, and Σ is the $m \times n$ matrix whose only nonzero entries are $\sigma_1, \dots, \sigma_r$ in positions $(1,1), \dots, (r,r)$.

The idea

V holds the input directions $\mathbf{v}_i$, and U holds the unit vectors in the directions of the outputs $A\mathbf{v}_i$, which are orthogonal by Thm. 9. Then $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ for every i, which is the matrix equation $AV = U\Sigma$.

Proof. 1. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be as in Thm. 9 and $V = [\,\mathbf{v}_1 \cdots \mathbf{v}_n\,]$, an orthogonal matrix.

2. For $i \leq r$, let $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$. By Thm. 9 these are orthonormal. Extend them to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_m$ of $\mathbb{R}^m$ (W3) and let $U = [\,\mathbf{u}_1 \cdots \mathbf{u}_m\,]$.

3. Then $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ for $i \leq r$ and $A\mathbf{v}_i = \mathbf{0}$ for $i > r$. Column by column this says

$$AV = [\,\sigma_1\mathbf{u}_1 \;\cdots\; \sigma_r\mathbf{u}_r \;\mathbf{0} \;\cdots\; \mathbf{0}\,] = U\Sigma.$$

4. Multiply on the right by $V^{-1} = V^\mathsf{T}$: $A = U\Sigma V^\mathsf{T}$.

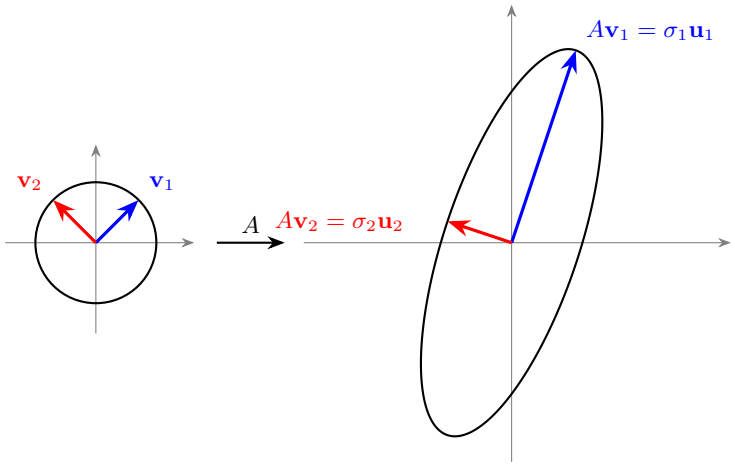
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The columns of U are **left singular vectors** and the columns of V are **right singular vectors**. Σ is determined by A, but U and V are not: for example, replacing $\mathbf{v}_i$ and $\mathbf{u}_i$ by $-\mathbf{v}_i$ and $-\mathbf{u}_i$ gives another SVD. Expanding $U\Sigma V^\mathsf{T}$ as a sum of (column)(row) products (W2) gives

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\mathsf{T} + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^\mathsf{T},$$

and transposing gives an SVD of $A^\mathsf{T} = V\Sigma^\mathsf{T} U^\mathsf{T}$ with the same singular values. **The question answered.** For the matrix A of the question:

1. The matrix
$$A^\mathsf{T}A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$
has characteristic polynomial $\lambda^2 - 50\lambda + 225 = (\lambda - 45)(\lambda - 5)$. For 45, the first row of $A^\mathsf{T}A - 45I$ is $(-20, 20)$, so $\mathbf{v}_1 = (1,1)/\sqrt{2}$; for 5, the perpendicular $\mathbf{v}_2 = (-1,1)/\sqrt{2}$ (Thm. 1). So $\sigma_1 = \sqrt{45} = 3\sqrt{5}$ and $\sigma_2 = \sqrt{5}$.
2. $A\mathbf{v}_1 = \frac{1}{\sqrt{2}}(3,9)$, of length $3\sqrt{5}$, so $\mathbf{u}_1 = \frac{1}{\sqrt{10}}(1,3)$. $A\mathbf{v}_2 = \frac{1}{\sqrt{2}}(-3,1)$, of length $\sqrt{5}$, so $\mathbf{u}_2 = \frac{1}{\sqrt{10}}(-3,1)$.
3. So
$$A = \underbrace{\frac{1}{\sqrt{10}}\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 3\sqrt{5} & 0 \\ 0 & \sqrt{5} \end{bmatrix}}_{\Sigma} \underbrace{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}}_{V^\mathsf{T}}.$$
A unit vector $\mathbf{x} = \cos\theta\, \mathbf{v}_1 + \sin\theta\, \mathbf{v}_2$ goes to $A\mathbf{x} = \sigma_1 \cos\theta\, \mathbf{u}_1 + \sigma_2 \sin\theta\, \mathbf{u}_2$, so the image of the circle is the ellipse with half-axes of lengths $3\sqrt{5}$ and $\sqrt{5}$ along $\mathbf{u}_1$ and $\mathbf{u}_2$, and $\|A\mathbf{x}\|$ is largest, $3\sqrt{5}$, at $\mathbf{x} = \pm\mathbf{v}_1$.



The matrix A of the question maps the unit circle to an ellipse. The perpendicular inputs $\mathbf{v}_1, \mathbf{v}_2$ go to the perpendicular half-axes $\sigma_1\mathbf{u}_1$, $\sigma_2\mathbf{u}_2$ of lengths $3\sqrt{5} \approx 6.7$ and $\sqrt{5} \approx 2.2$. The right picture is drawn at half the scale of the left one.

Example (a 3×2 matrix: extending U). Find an SVD of

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

1. The matrix
$$A^\mathsf{T}A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$
has characteristic polynomial $\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$, with unit eigenvectors $\mathbf{v}_1 = (1,1)/\sqrt{2}$ for 3 and $\mathbf{v}_2 = (1,-1)/\sqrt{2}$ for 1. So $\sigma_1 = \sqrt{3}$ and $\sigma_2 = 1$.
2. $A\mathbf{v}_1 = \frac{1}{\sqrt{2}}(2,1,1)$, so $\mathbf{u}_1 = \frac{1}{\sqrt{6}}(2,1,1)$; $A\mathbf{v}_2 = \frac{1}{\sqrt{2}}(0,-1,1)$, so $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(0,-1,1)$.
3. $\mathbf{u}_3$ must be a unit vector orthogonal to $\mathbf{u}_1, \mathbf{u}_2$, that is, a unit vector in $(\operatorname{Col} A)^\perp = \operatorname{Nul} A^\mathsf{T}$ (W5). Solving $2x_1 + x_2 + x_3 = 0$, $-x_2 + x_3 = 0$ gives $\mathbf{u}_3 = \frac{1}{\sqrt{3}}(1,-1,-1)$.

4. So
$$A = \begin{bmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

What the SVD gives. Let $A = U\Sigma V^\mathsf{T}$ have rank r.

- **Bases of the four subspaces.** $\mathbf{u}_1, \ldots, \mathbf{u}_r$ is an orthonormal basis of $\operatorname{Col} A$ (Thm. 9). The remaining $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$ are $m - r$ orthonormal vectors orthogonal to $\operatorname{Col} A$, and $\dim(\operatorname{Col} A)^\perp = m - r$ (W5), so they are an orthonormal basis of $(\operatorname{Col} A)^\perp = \operatorname{Nul} A^\mathsf{T}$. The vectors $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$ satisfy $A\mathbf{v}_i = \mathbf{0}$ and there are $n - r = \dim \operatorname{Nul} A$ of them, so they are an orthonormal basis of $\operatorname{Nul} A$. Then $\mathbf{v}_1, \ldots, \mathbf{v}_r$ is an orthonormal basis of $(\operatorname{Nul} A)^\perp$, which is $\operatorname{Row} A$ because $(\operatorname{Row} A)^\perp = \operatorname{Nul} A$ and $(W^\perp)^\perp = W$ (W5).
- **Reduced SVD and pseudoinverse.** Keep only the first r columns: $U_r = [\,\mathbf{u}_1 \cdots \mathbf{u}_r\,]$, $V_r = [\,\mathbf{v}_1 \cdots \mathbf{v}_r\,]$ and $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$. The zero rows and columns of Σ contribute nothing to the product, so $A = U_r D V_r^\mathsf{T}$. The **pseudoinverse** is $A^+ = V_r D^{-1} U_r^\mathsf{T}$. Then $\hat{\mathbf{x}} = A^+\mathbf{b}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$, because $V_r^\mathsf{T} V_r = I$ (W2) and
$$A\hat{\mathbf{x}} = U_r D V_r^\mathsf{T} V_r D^{-1} U_r^\mathsf{T}\mathbf{b} = U_r U_r^\mathsf{T}\mathbf{b} = \operatorname{proj}_{\operatorname{Col} A}\mathbf{b} \quad \text{(W6)}.$$
  For the 3×2 example ($r = 2$),
$$V_r D^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{bmatrix}, \qquad U_r^\mathsf{T} = \begin{bmatrix} 2/\sqrt{6} & 1/\sqrt{6} & 1/\sqrt{6} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$
  so the rows of $A^+ = V_r D^{-1} U_r^\mathsf{T}$ are
$$\frac{1}{6}(2,1,1) + \frac{1}{2}(0,-1,1) = \frac{1}{3}(1,-1,2), \qquad \frac{1}{6}(2,1,1) - \frac{1}{2}(0,-1,1) = \frac{1}{3}(1,2,-1).$$
- **Invertible matrix theorem, concluded.** For an $n \times n$ matrix A, each of these is equivalent to invertibility: $(\operatorname{Col} A)^\perp = \{\mathbf{0}\}$; $(\operatorname{Nul} A)^\perp = \mathbb{R}^n$; $\operatorname{Row} A = \mathbb{R}^n$; A has n nonzero singular values. Each says $\operatorname{rank} A = n$: the first because $\dim \operatorname{Col} A = n - \dim(\operatorname{Col} A)^\perp$; the second because $(\operatorname{Nul} A)^\perp = \mathbb{R}^n$ exactly when $\operatorname{Nul} A = \{\mathbf{0}\}$ (take complements, W5); the third because $\dim \operatorname{Row} A = \operatorname{rank} A$; the last by Thm. 9.
- **Condition number.** For invertible A, σ_1/σ_n is the **condition number**: the ratio of the largest to the smallest stretch. Lay notes (without proof here) that a large condition number means the solution of $A\mathbf{x} = \mathbf{b}$ can change a lot when $\mathbf{b}$ changes a little.

Shortcuts

- $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$: no need to diagonalize AA^T. Only the extra columns of U (beyond r) need a separate computation, from $\operatorname{Nul} A^\mathsf{T}$. Reason: step 2 of the proof. Instance: the 3×2 example.
- **Use the smaller of $A^\mathsf{T}A$ and AA^T**: they have the same nonzero eigenvalues. Reason: if $A^\mathsf{T}A\mathbf{v} = \lambda\mathbf{v}$ with $\lambda \neq 0$, then $A\mathbf{v} \neq \mathbf{0}$ and $AA^\mathsf{T}(A\mathbf{v}) = \lambda A\mathbf{v}$. If you diagonalize AA^T instead, get the $\mathbf{v}_i$ from $\mathbf{v}_i = A^\mathsf{T}\mathbf{u}_i/\sigma_i$ (because $A^\mathsf{T}\mathbf{u}_i = A^\mathsf{T}A\mathbf{v}_i/\sigma_i = \sigma_i\mathbf{v}_i$). Instance: for the 2×3 matrix A^T of the 3×2 example, $(A^\mathsf{T})^\mathsf{T}A^\mathsf{T} = AA^\mathsf{T}$ is 3×3, but $A^\mathsf{T}(A^\mathsf{T})^\mathsf{T} = A^\mathsf{T}A$ is the 2×2 matrix above; its SVD is $V\Sigma^\mathsf{T}U^\mathsf{T}$.
- **Check**: $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$ for each i, that is, $AV = U\Sigma$. Instance: in the 2×2 example, $\sigma_1\mathbf{u}_1 = (3\sqrt{5}/\sqrt{10})(1,3) = (1/\sqrt{2})(3,9) = A\mathbf{v}_1$.
- **Rank**: the number of nonzero singular values (Thm. 9). Instance: both examples have rank 2.

Traps

- Singular values are not eigenvalues of A. For $A = \operatorname{diag}(-2, 1)$, the eigenvalues are $-2, 1$ and the singular values are $2, 1$. For
$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$
  the eigenvalues are $0, 0$, but $A^\mathsf{T}A = \operatorname{diag}(0, 1)$, so the singular values are $1, 0$.

- Do not compute U and V from AA^{T} and $A^{\mathsf{T}}A$ separately: the signs may not match. For $A = \operatorname{diag}(2, 1)$, the choices $\mathbf{v}_1 = \mathbf{e}_1$, $\mathbf{u}_1 = -\mathbf{e}_1$ are eigenvectors of the right matrices, but they give $U\Sigma V^{\mathsf{T}}$ with $(1, 1)$ entry -2, not A.
- List the singular values in decreasing order, with the columns of V and U in the matching order. For $A = \operatorname{diag}(1, 2)$, $U = V = I$ with $\Sigma = \operatorname{diag}(1, 2)$ multiplies out to A but is not the SVD: $\sigma_1 = 2$, with $\mathbf{v}_1 = \mathbf{u}_1 = \mathbf{e}_2$.

§2.3 Homogeneous second-order linear equations (Nagle 4.2)

The question. Find all functions $y(t)$ with $y'' + y' - 6y = 0$, and the one with $y(0) = 2$, $y'(0) = -1$. How many conditions does it take to single out one solution, and how do we know we have found all of them?

We study $ay'' + by' + cy = 0$ with constants $a \neq 0$, b, c. A solution's second derivative is a combination of its first and zeroth derivatives, and derivatives of e^{rt} are multiples of e^{rt}, so try $y = e^{rt}$:

$$ar^2e^{rt} + bre^{rt} + ce^{rt} = e^{rt}(ar^2 + br + c).$$

Since $e^{rt} \neq 0$, $y = e^{rt}$ is a solution exactly when $ar^2 + br + c = 0$, the **auxiliary equation** (characteristic equation).

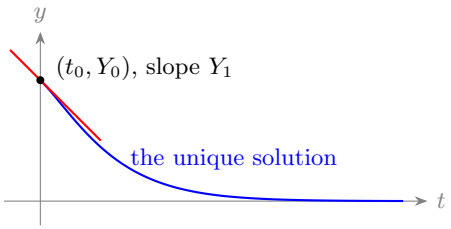
Combinations of solutions are solutions. If y_1, y_2 are solutions, then

$$a(c_1y_1 + c_2y_2)'' + b(c_1y_1 + c_2y_2)' + c(c_1y_1 + c_2y_2) = c_1(ay_1'' + by_1' + cy_1) + c_2(ay_2'' + by_2' + cy_2) = 0.$$

In linear-algebra terms, $L(y) = ay'' + by' + cy$ is a linear map on twice-differentiable functions, and the solutions are its null space, a subspace.

Theorem (Existence and uniqueness (Nagle Thm. 1))
For real $a \neq 0$, b, c, t_0, Y_0, Y_1, the problem $ay'' + by' + cy = 0$, $y(t_0) = Y_0$, $y'(t_0) = Y_1$ has exactly one solution, valid for all t.

Nagle proves this theorem in a later chapter; we use it as stated (cited). In particular, a solution with $y(t_0) = y'(t_0) = 0$ is identically 0.



A value and a slope at one time t_0 determine the solution. Here $y'' + 6y' + 9y = 0$ with $y(0) = 1$, $y'(0) = -1$ has the single solution $e^{-3t}(1 + 2t)$ (substitute it to check); the red segment is the prescribed slope.

Definition. Two functions y_1, y_2 are **linearly independent** on an interval I if neither is a constant multiple of the other on all of I; otherwise they are **linearly dependent**. Equivalently, they are dependent exactly when $c_1y_1 + c_2y_2 = 0$ on all of I for some constants c_1, c_2 not both 0 (if $c_1 \neq 0$, then $y_1 = -(c_2/c_1)y_2$). The **Wronskian** of y_1, y_2 is

$$W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t) = \det \begin{bmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{bmatrix}.$$

Theorem (A zero Wronskian means dependence (Nagle Lemma 1))

If y_1, y_2 are solutions of $ay'' + by' + cy = 0$ on $(-\infty, \infty)$ and $W[y_1, y_2](\tau) = 0$ at one point τ, then y_1 and y_2 are linearly dependent.

The idea

$W(\tau) = 0$ says the vectors $(y_1(\tau), y_1'(\tau))$ and $(y_2(\tau), y_2'(\tau))$ are dependent. So some multiple ky_1 has the same value and slope at τ as y_2; by uniqueness, $ky_1 = y_2$ everywhere.

Proof. 1. If $y_1(\tau) \neq 0$, let $k = y_2(\tau)/y_1(\tau)$. Then ky_1 is a solution whose value at τ is $y_2(\tau)$. Its slope at τ is

$$ky_1'(\tau) = \frac{y_2(\tau)y_1'(\tau)}{y_1(\tau)} = y_2'(\tau),$$

using $W(\tau) = 0$. By uniqueness $y_2 = ky_1$.

2. If $y_1(\tau) = 0$ but $y_1'(\tau) \neq 0$, then $W(\tau) = 0$ gives $y_2(\tau) = 0$. Let $k = y_2'(\tau)/y_1'(\tau)$; then ky_1 and y_2 both have value 0 and slope $y_2'(\tau)$ at τ, so $y_2 = ky_1$ by uniqueness.

3. If $y_1(\tau) = y_1'(\tau) = 0$, then $y_1 \equiv 0$ by uniqueness, and $y_1 = 0 \cdot y_2$.

□

Theorem (General solution (Nagle Thm. 2))

If y_1, y_2 are linearly independent solutions of $ay'' + by' + cy = 0$ on $(-\infty, \infty)$, then for every t_0, Y_0, Y_1 there are unique constants with $c_1y_1 + c_2y_2$ solving the initial value problem. Hence every solution is $c_1y_1 + c_2y_2$: $\{y_1, y_2\}$ is a basis of the solution space, which has dimension 2.

Proof. 1. The conditions are the linear system

$$\begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} Y_0 \\ Y_1 \end{bmatrix},$$

whose determinant is $W(t_0)$. If $W(t_0)$ were 0, Nagle Lemma 1 would make y_1, y_2 dependent; so $W(t_0) \neq 0$. So there is exactly one solution (c_1, c_2).

2. If y is any solution, choose c_1, c_2 so that $c_1y_1 + c_2y_2$ has the value and slope of y at t_0. By uniqueness, $y = c_1y_1 + c_2y_2$.

□

So for two solutions, the Wronskian is either never 0 (independent) or always 0 (dependent).

Distinct real roots. If $ar^2 + br + c = 0$ has real roots $r_1 \neq r_2$, then $e^{r_1 t}$ and $e^{r_2 t}$ are solutions, independent because their ratio $e^{(r_1 - r_2)t}$ is not constant. The general solution is $c_1e^{r_1 t} + c_2e^{r_2 t}$.

The question answered. $r^2 + r - 6 = (r - 2)(r + 3)$, so the general solution is $y = c_1e^{2t} + c_2e^{-3t}$. The conditions give $c_1 + c_2 = 2$ and $2c_1 - 3c_2 = -1$, so $c_1 = c_2 = 1$:

$$y = e^{2t} + e^{-3t}.$$

Two conditions (value and slope at one time) single out one solution, and the theorems above show there are no other solutions.

Irrational roots are handled the same way. For $y'' + 4y' + y = 0$, the roots are

$$r = \frac{-4 \pm \sqrt{12}}{2} = -2 \pm \sqrt{3},$$

so $y = c_1e^{(-2+\sqrt{3})t} + c_2e^{(-2-\sqrt{3})t}$. With $y(0) = 0$, $y'(0) = 1$: $c_1 + c_2 = 0$ and $(-2 + \sqrt{3})c_1 + (-2 - \sqrt{3})c_2 = 1$, so $2\sqrt{3}\,c_1 = 1$ and

$$y = \frac{1}{2\sqrt{3}}\left(e^{(-2+\sqrt{3})t} - e^{(-2-\sqrt{3})t}\right).$$

Conditions at $t_0 \neq 0$. Write the general solution in powers of $t - t_0$: $e^{r(t-t_0)} = e^{-rt_0}e^{rt}$ is a constant multiple of e^{rt}, so it is still a solution, and at $t = t_0$ it equals 1. Example: $y'' - y' - 2y = 0$, $y(1) = 0$, $y'(1) = 3$. The roots are 2 and -1, so take $y = c_1e^{2(t-1)} + c_2e^{-(t-1)}$. Then $c_1 + c_2 = 0$ and $2c_1 - c_2 = 3$, so $c_1 = 1$, $c_2 = -1$:

$$y = e^{2(t-1)} - e^{-(t-1)}.$$

Repeated root. If $b^2 - 4ac = 0$, the only root is $r = -b/(2a)$, so $2ar + b = 0$. Then te^{rt} is a second solution: with $y = te^{rt}$, $y' = e^{rt} + rte^{rt}$ and $y'' = 2re^{rt} + r^2te^{rt}$, so

$$ay'' + by' + cy = (2ar + b)e^{rt} + (ar^2 + br + c)te^{rt} = 0.$$

e^{rt} and te^{rt} are independent (their ratio t is not constant), so the general solution is $c_1e^{rt} + c_2te^{rt}$.

Example. $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = -1$. Here $r^2 + 6r + 9 = (r + 3)^2$, so $y = (c_1 + c_2t)e^{-3t}$ and $y' = (c_2 - 3c_1 - 3c_2t)e^{-3t}$. Then $c_1 = 1$ and $c_2 - 3c_1 = -1$, so $c_2 = 2$ and

$$y = (1 + 2t)e^{-3t}.$$

Higher order. The same substitution works for any order: each root r of the auxiliary polynomial gives e^{rt}. Three functions are linearly independent if none is a combination of the other two, and an nth-order equation needs n independent solutions (Nagle Ch. 6, cited). For $y''' - y = 0$, $r^3 - 1 = (r - 1)(r^2 + r + 1)$ has the real root 1, and the complex roots $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ are handled in the next section.

Shortcuts

- **Initial conditions at $t_0 = 0$, distinct roots**: the system is $c_1 + c_2 = y(0)$, $r_1c_1 + r_2c_2 = y'(0)$. Instance: the question above. At $t_0 \neq 0$, use $e^{r(t-t_0)}$ and the same system.
- **Repeated root at $t_0 = 0$**: $c_1 = y(0)$ and $c_2 = y'(0) - r\,y(0)$, because $y'(0) = rc_1 + c_2$. Instance: $c_2 = -1 - (-3)(1) = 2$ above.
- **Independence of two solutions**: compute the Wronskian at one convenient point; nonzero there means independent, zero there means dependent. Reason: Nagle Lemma 1 and the remark after Nagle Thm. 2. Instance: $W[e^{2t}, e^{-3t}](0) = 1 \cdot (-3) - 2 \cdot 1 = -5 \neq 0$.
- **First-order equations**: $ay' + by = 0$ has auxiliary equation $ar + b = 0$, so $y = ce^{-bt/a}$. Reason: the substitution $y = e^{rt}$ gives $(ar + b)e^{rt}$. Instance: $2y' - 6y = 0$ gives $y = ce^{3t}$.

Traps

- A repeated root gives only one exponential; $c_1e^{rt} + c_2e^{rt} = (c_1 + c_2)e^{rt}$ has one free constant and cannot meet two conditions. The second solution is te^{rt}.
- The Wronskian test for dependence is for two *solutions* of one equation. For arbitrary functions it fails: $y_1 = t$, $y_2 = t^2$ have $W = t \cdot 2t - 1 \cdot t^2 = t^2$, which is 0 at $t = 0$, but t^2 is not a constant multiple of t.
- Uniqueness is for a value and a slope at the *same* point. Values at two points (boundary conditions) need not determine a solution: $y'' + y = 0$ with $y(0) = 0$, $y(\pi) = 0$ is solved by $c \sin t$ for every c (check: $(\sin t)'' = -\sin t$).

§2.4 Auxiliary equations with complex roots (Nagle 4.3)

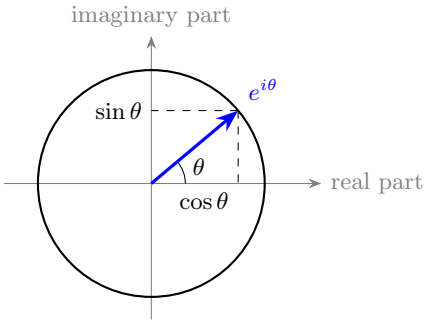
The question. $y'' + y = 0$ is solved by $\cos t$ and $\sin t$, but its auxiliary equation $r^2 + 1 = 0$ has the roots $\pm i$. How do complex roots produce real solutions? Find the solution of $y'' + 4y' + 13y = 0$ with $y(0) = 2$, $y'(0) = -1$.

Euler's formula. If the Maclaurin series of e^x is used for $x = i\theta$, then, since $i^2 = -1$, the even powers give the series of $\cos\theta$ and the odd powers give i times the series of $\sin\theta$:

$$e^{i\theta} = \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\Big) + i\Big(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\Big) = \cos\theta + i\sin\theta.$$

With the law of exponents, for real α, β this defines

$$e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos\beta t + i\sin\beta t).$$



Euler's formula: $e^{i\theta} = \cos\theta + i\sin\theta$ is the point at angle θ on the unit circle of the complex plane. Multiplying by $e^{\alpha t}$ scales it; as t grows, $e^{(\alpha+i\beta)t}$ turns at angular speed β.

The derivative rule still holds. By the product rule,

$$\begin{aligned}\frac{d}{dt}\big[e^{\alpha t}(\cos\beta t + i\sin\beta t)\big] &= e^{\alpha t}\big[\alpha(\cos\beta t + i\sin\beta t) + \beta(-\sin\beta t + i\cos\beta t)\big] \\ &= (\alpha + i\beta)\, e^{(\alpha+i\beta)t},\end{aligned}$$

because $i\beta(\cos\beta t + i\sin\beta t) = \beta(-\sin\beta t + i\cos\beta t)$. So the substitution of e^{rt} works for complex r too: if $ar^2 + br + c = 0$ with $b^2 - 4ac < 0$, the roots are $\alpha \pm i\beta$ with

$$\alpha = -\frac{b}{2a}, \qquad \beta = \frac{\sqrt{4ac - b^2}}{2a},$$

and $e^{(\alpha+i\beta)t}$ is a complex-valued solution.

Theorem (Real solutions from a complex solution (Nagle Lemma 2))
If $z(t) = u(t) + iv(t)$ solves $ay'' + by' + cy = 0$ with a, b, c real, then the real part u and the imaginary part v are real solutions.

Proof. Since $z'' = u'' + iv''$ and $z' = u' + iv'$,

$$0 = az'' + bz' + cz = (au'' + bu' + cu) + i(av'' + bv' + cv),$$

and both brackets are real because a, b, c are real. A complex number is 0 only if its real and imaginary parts are 0 (W4), so both brackets are 0. □

Complex conjugate roots $\alpha \pm i\beta$. Applying the lemma to $e^{(\alpha+i\beta)t} = e^{\alpha t}\cos\beta t + ie^{\alpha t}\sin\beta t$ gives the real solutions $e^{\alpha t}\cos\beta t$ and $e^{\alpha t}\sin\beta t$. At $t = 0$ their Wronskian is $1 \cdot \beta - \alpha \cdot 0 = \beta \neq 0$, so they are independent, and the general solution is

$$y = e^{\alpha t}(c_1\cos\beta t + c_2\sin\beta t).$$

The root $\alpha - i\beta$ gives the same two real functions (up to sign), so it adds nothing new. The complex form $d_1e^{(\alpha+i\beta)t} + d_2e^{(\alpha-i\beta)t}$ with complex constants is also a general solution; Euler's formula turns it into the real form above (Nagle Example 1).

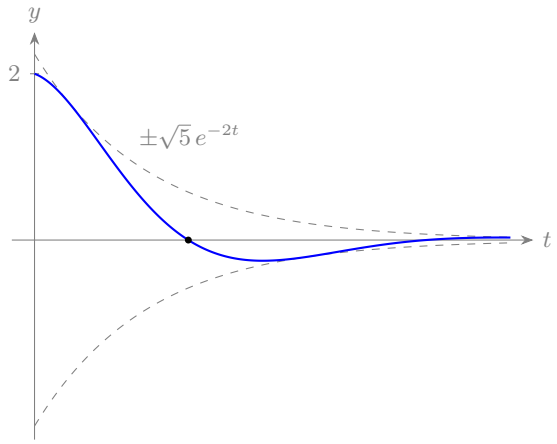
The question answered. The roots of $r^2 + 4r + 13 = 0$ are

$$r = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i,$$

so $y = e^{-2t}(c_1 \cos 3t + c_2 \sin 3t)$. Then $y(0) = c_1 = 2$, and

$$y'(0) = -2c_1 + 3c_2 = -1, \qquad \text{so } c_2 = 1, \qquad y = e^{-2t}(2\cos 3t + \sin 3t).$$

The first time with $y = 0$ is when $2\cos 3t + \sin 3t = 0$, that is, $\tan 3t = -2$ with $\cos 3t < 0 < \sin 3t$: $3t = \pi - \arctan 2$, so $t \approx 0.68$.

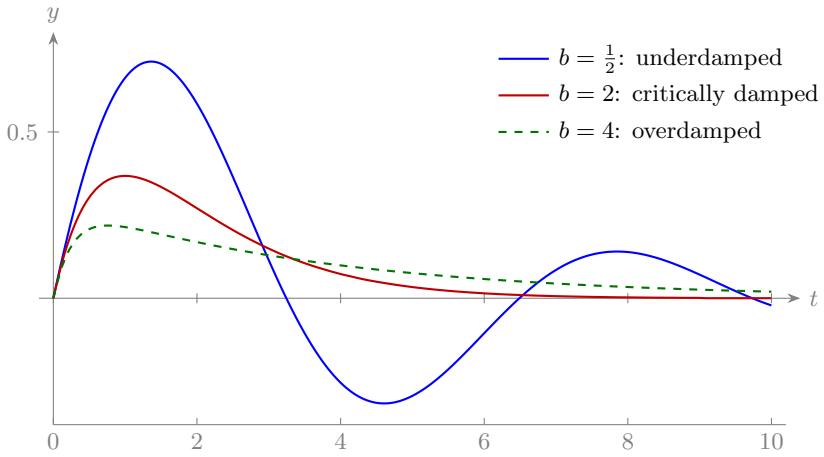


$y = e^{-2t}(2\cos 3t + \sin 3t)$ oscillates with angular frequency $\beta = 3$ inside the decaying envelope $\pm\sqrt{c_1^2 + c_2^2}\,e^{\alpha t} = \pm\sqrt{5}\,e^{-2t}$ (dashed). The dot is the first zero, $t \approx 0.68$.

The mass–spring model (W8): $my'' + by' + ky = 0$ with $m > 0$. The roots satisfy $r_1 + r_2 = -b/m$ and $r_1 r_2 = k/m$.

- **Oscillation** happens exactly when $b^2 < 4mk$ (complex roots), with decay rate $\alpha = -b/(2m)$, angular frequency $\beta = \sqrt{4mk - b^2}/(2m)$ and frequency $\beta/(2\pi)$ oscillations per unit time. With $b = 0$ the motion is $c_1 \cos \beta t + c_2 \sin \beta t$, with no decay.
- **No oscillation when** $b^2 \geq 4mk$: a solution $c_1 e^{r_1 t} + c_2 e^{r_2 t}$ (not identically 0) is 0 only where $e^{(r_1 - r_2)t} = -c_2/c_1$, which happens at most once because the exponential is strictly monotone; for a repeated root, $(c_1 + c_2 t)e^{rt} = 0$ only at $t = -c_1/c_2$. So the mass crosses equilibrium at most once.
- **Long-run behaviour.** If $b > 0$ and $k > 0$, every solution tends to 0: complex roots have real part $-b/(2m) < 0$; real roots have positive product and negative sum, so both are negative. If $k < 0$, the product $r_1 r_2$ is negative, so one root is positive, and solutions with a nonzero coefficient on it grow without bound. Instance: $y'' + y' - 6y = 0$ (here $k = -6$) has the solution $e^{2t} + e^{-3t} \to \infty$ found above. If $b < 0 < k$, the real parts are positive, and the solutions grow: the damping adds energy (Nagle Example 4).
- **The three damping cases** ($m, k > 0$, $b > 0$; with $b = 0$ the motion is undamped, first bullet), named by the sign of $b^2 - 4mk$:
  - **underdamped**, $b^2 < 4mk$: complex roots $\alpha \pm i\beta$ with $\alpha = -b/(2m)$; the mass oscillates inside the envelope $\pm\sqrt{c_1^2 + c_2^2}\,e^{\alpha t}$;
  - **critically damped**, $b^2 = 4mk$: the repeated root $r = -b/(2m)$, so $y = (c_1 + c_2 t)e^{rt}$;
  - **overdamped**, $b^2 > 4mk$: two different real roots, both negative (long-run bullet above), so $y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$.

  In the last two cases the mass crosses equilibrium at most once (the bullet above). The figure shows all three for $m = k = 1$, started at equilibrium with velocity 1: $y(0) = 0$, $y'(0) = 1$.



$y'' + by' + y = 0$ with $y(0) = 0$, $y'(0) = 1$. Underdamped (blue): $\frac{4}{\sqrt{15}} e^{-t/4} \sin \frac{\sqrt{15}}{4} t$ crosses 0 again and again. Critically damped (red): te^{-t}. Overdamped (dashed): $\frac{1}{2\sqrt{3}} \left(e^{(-2+\sqrt{3})t} - e^{(-2-\sqrt{3})t}\right)$, slowed by its root $-2 + \sqrt{3} \approx -0.27$ near 0. Neither of the last two crosses 0 for $t > 0$.

The same equation for an electric circuit. In a series circuit with an inductor (inductance L), a resistor (resistance R) and a capacitor (capacitance C) and no voltage source, let $q(t)$ be the charge on the capacitor, so the current is $I = q'$. The voltage drops across the three parts are $LI' = Lq''$, $RI = Rq'$ and q/C (these three laws are cited from physics), and by Kirchhoff's voltage law (cited) they add up to 0:

$$Lq'' + Rq' + \frac{1}{C} q = 0.$$

This is $my'' + by' + ky = 0$ with L in place of m, R in place of b and $1/C$ in place of k, so everything above carries over: the charge oscillates exactly when $R^2 < 4L/C$, with decay rate $-R/(2L)$ and angular frequency $\sqrt{4L/C - R^2}/(2L)$. Instance: $L = 1$, $R = 2$, $C = \frac{1}{5}$ give

$$q'' + 2q' + 5q = 0, \qquad r = -1 \pm 2i,$$

and here $R^2 = 4 < 20 = 4L/C$, so $q = e^{-t}(c_1 \cos 2t + c_2 \sin 2t)$.

Shortcuts

- **Read α, β from the coefficients**: $\alpha = -b/(2a)$, and β is the square root of $4ac - b^2$ divided by $2a$; no complex arithmetic is needed. Instance: $a = 1$, $b = 4$, $c = 13$ give $\alpha = -2$, $4ac - b^2 = 36$ and $\beta = 6/2 = 3$.
- **Initial conditions at $t_0 = 0$**: $c_1 = y(0)$ and $c_2 = (y'(0) - \alpha\, y(0))/\beta$, because $y'(0) = \alpha c_1 + \beta c_2$. Instance: $c_2 = (-1 + 2 \cdot 2)/3 = 1$ above.
- **Check the roots**: $\alpha \pm i\beta$ are the roots of $r^2 - 2\alpha r + (\alpha^2 + \beta^2) = 0$ (expand $(r - \alpha)^2 + \beta^2$), which must be the auxiliary equation divided by a. Instance: $\alpha = -2$, $\beta = 3$ give $r^2 + 4r + 13$.
- **Amplitude and phase**: $c_1 \cos \beta t + c_2 \sin \beta t = R \cos(\beta t - \varphi)$ with $R = \sqrt{c_1^2 + c_2^2}$, $\cos \varphi = c_1/R$, $\sin \varphi = c_2/R$, by the formula for $\cos(x - y)$ (W7). So $|y| \le Re^{\alpha t}$, the envelope in the figure. Instance: $R = \sqrt{4 + 1} = \sqrt{5}$ above.
- **Which damping case**: compare b^2 with $4mk$ (for a circuit, R^2 with $4L/C$); no roots needed. Reason: the roots are $(-b \pm \sqrt{b^2 - 4mk})/(2m)$, so the sign of $b^2 - 4mk$ decides between complex, repeated and distinct real roots. Condition: $m, k > 0$, $b > 0$. Instance: $y'' + 4y' + y = 0$ has $16 > 4$: overdamped.
- **Higher order**: factor the auxiliary polynomial; each real root r gives e^{rt} and each pair $\alpha \pm i\beta$ gives $e^{\alpha t} \cos \beta t$, $e^{\alpha t} \sin \beta t$. Instance: $y''' - y = 0$ has the general solution

$$y = c_1 e^t + e^{-t/2} \left(c_2 \cos \frac{\sqrt{3}}{2} t + c_3 \sin \frac{\sqrt{3}}{2} t\right).$$

Traps

- Divide by $2a$ in both α and β. For $9y'' + 6y' + 10y = 0$, $4ac - b^2 = 360 - 36 = 324 = 18^2$, so $\beta = 18/18 = 1$ (not 18) and $\alpha = -6/18 = -\frac{1}{3}$.
- Both $\cos \beta t$ and $\sin \beta t$ are needed: $c_1 e^{\alpha t} \cos \beta t$ alone has one constant and cannot meet two conditions.
- The angular frequency is β, not α; the sign of α decides decay ($\alpha < 0$) or growth ($\alpha > 0$). In the example, the zeros are $\pi/3$ apart, matching $\beta = 3$; $\alpha = -2$ only sets how fast the envelope shrinks.

§3 Homework this week

HW11, Nagle Section 4.3: 3, 8, 15, 17, 24, 25, 28, 31(a,b,d), 34.

HW10, Lay Section 7.1:

- **#4, #6** (symmetric?). $A^\mathsf{T} = A$: compare a_{ij} with a_{ji}, watching the signs.
- **#11** (orthogonal? inverse). Test for orthonormal columns (Week 9); if orthogonal, $U^{-1} = U^\mathsf{T}$.
- **#18** (orthogonally diagonalize, eigenvalues given). The eigenvalues are distinct, so the eigenvectors are already orthogonal (Lay Thm. 1); only normalize.
- **#22** (4×4, two eigenvalues given). Each eigenspace has dimension 2 (spectral theorem (b)); Gram–Schmidt only inside an eigenspace, then normalize.
- **#23, #24** (verify eigenvectors, then diagonalize). $A\mathbf{v}$ gives the eigenvalue; Last eigenspace by orthogonality, as in the repeated-eigenvalue example.
- **#25, #28** (true/false: PDP^T is symmetric). Transpose: $(PDP^\mathsf{T})^\mathsf{T} = PDP^\mathsf{T}$.
- **#30, #32** (true/false). Lay Thm. 1; spectral theorem (b).

Lay Section 7.4:

- **#1, #3** (singular values). Eigenvalues of $A^\mathsf{T}A$, square roots, decreasing order; signs disappear, so singular values are not eigenvalues.
- **#7** (SVD, 2×2). Unit eigenvectors of $A^\mathsf{T}A$ give V; $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$; check $AV = U\Sigma$.
- **#9** (SVD, 3×2). As in the 3×2 example: the third column of U is a unit vector in $\operatorname{Nul} A^\mathsf{T}$.
- **#14** (unit $\mathbf{x}$ maximizing $\|A\mathbf{x}\|$). The largest stretch is σ_1, reached at $\mathbf{v}_1$ from #7.

HW10, Nagle Section 4.2:

- 1: factor the auxiliary equation; distinct real roots.
- 9: the auxiliary polynomial is a perfect square; repeated root, so use e^{rt} and te^{rt}.
- 14: one root is 0, and $e^{0t} = 1$ is a solution.
- 20: repeated root with conditions at $t_0 = 1$; write the solution in powers of $t - 1$.
- 21: the first-order shortcut, $ar + b = 0$.
- 28, 30, 31: Definition of dependence; check whether the ratio y_2/y_1 is constant on the interval (for 31, simplify with $\sec^2 t = 1 + \tan^2 t$).

§4 Problem set

P1. Orthogonally diagonalize

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}.$$

P2. Orthogonally diagonalize

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

P3. Write the spectral decomposition of the matrix in P1 and check that it adds up to A.

P4. Let A be symmetric with $A^2 = 0$. Show that $A = 0$.

P5. Find a singular value decomposition of

$$A = \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix}.$$

P6. Find a singular value decomposition of

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix},$$

and use it to give orthonormal bases of $\operatorname{Col} A$, $\operatorname{Nul} A^\mathsf{T}$, $\operatorname{Row} A$ and $\operatorname{Nul} A$.

P7. Solve $y'' - 3y' - 10y = 0$ with $y(0) = 0$, $y'(0) = 7$.

P8. Solve $4y'' + 12y' + 9y = 0$ with $y(0) = 2$, $y'(0) = 0$.

P9. Solve $y'' + 2y' + 10y = 0$ with $y(0) = 0$, $y'(0) = 6$.

P10. Compute the Wronskian of $e^{\alpha t}\cos\beta t$ and $e^{\alpha t}\sin\beta t$, and conclude that they are independent when $\beta \neq 0$.